

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – CODING CHALLENGE

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 3 – Coding Challenge
Weightage:	30% of CIA	Slot:	B
CO5 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	G.V.S.HEMANTH	Reg. No.:	192425186
Section / Batch:	AI & DS	Date:	27-07-2026

Code of Conduct

I ____ G.V.S.HEMANTH(192425186)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ G.V.S.HEMANTH _____

Instructions

- Develop a complete and modular Python program that meets all the functional requirements specified in the coding challenge using appropriate algorithms, data structures, and programming practices.
- Test the program using multiple valid and invalid test cases, and clearly display the input, processing, and output to verify the correctness of the implementation.
- Follow standard coding practices by using meaningful variable names, proper indentation, comments, exception handling (where applicable), and upload the source code, sample outputs, and README file to the designated GitHub repository.
- Duration: 40 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Matching	Regular Expressions, Basic Regular Expression Patterns	1
Finite State Automata Implementation	Finite State Automata, Models and Algorithms	2
Advanced Regular Expression Search	Regular Expressions, Basic Regular Expression Patterns	3

Section 1: Intelligent Email and Password Validator using Regular Expressions

1. Problem Statement:

A software company is developing a user registration system. Before creating an account, the system must validate user credentials based on predefined security policies.

Develop a Python application that validates:

Email Address

Password

Mobile Number

using Regular Expressions.

Validation Rules

Email

Must start with an alphabet.

May contain letters, digits, '.', '_' before '@'.

Domain should contain only alphabets.

Valid extensions: .com, .org, .edu, .net, .in

Password

Minimum 8 characters

At least one uppercase letter

At least one lowercase letter

At least one digit

At least one special character (@, #, \$, %, &, !)

Mobile Number

Must contain exactly 10 digits.

First digit should be between 6–9.

Expected Output

Display:

Valid Email / Invalid Email

Strong Password / Weak Password

Valid Mobile Number / Invalid Mobile Number

Section 2: Design a Finite State Automata (FSA) Simulator

2. Problem Statement

Develop a Python-based Deterministic Finite Automata (DFA) simulator.

The program should:

- Accept the DFA description
 - States
 - Input alphabet
 - Transition table
 - Initial state
 - Final state(s)
- Accept multiple input strings.
- Simulate state transitions.
- Display whether the string is Accepted or Rejected.

Example

Language:

Strings ending with "ab"

Input

abaab

Output

Transition Path:

$q_0 \rightarrow q_1 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2$

Accepted

Section 3: Smart Pattern Matching Engine using Regular Expressions**3. Problem Statement**

Develop a Python-based text search engine that performs pattern matching using Regular Expressions.

The system should support:

Word Search

Prefix Search

Suffix Search

Date Search

Phone Number Search

Hashtag Search

Mention (@username) Search

Example Input

Meeting on 12/09/2026

Call 9876543210

#NLP

@OpenAI

natural language processing

User Menu

1.Search Date

2.Search Phone Number

3.Search Hashtag

4.Search Mention

5.Search Prefix

6.Search Suffix

Expected Output

Display all matching patterns based on user selection.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Q. No. / Criteria	Problem Understanding	Algorithm	Code Implementation	Competitive Performance	Test Case Handling	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1.

```
import re

# Function to validate email
def validate_email(email):
    pattern = r'^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$'
    if re.fullmatch(pattern, email):
        return "Valid Email"
    else:
        return "Invalid Email"

# Function to validate password
def validate_password(password):
    pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%&!])[A-Za-z\d@#%&!]{8,}$'
    if re.fullmatch(pattern, password):
        return "Strong Password"
    else:
        return "Weak Password"

# Function to validate mobile number
def validate_mobile(mobile):
    pattern = r'^[6-9]\d{9}$'
    if re.fullmatch(pattern, mobile):
        return "Valid Mobile Number"
    else:
        return "Invalid Mobile Number"

# User Input
email = input("Enter Email Address: ")
password = input("Enter Password: ")
mobile = input("Enter Mobile Number: ")

# Display Results
print(validate_email(email))
print(validate_password(password))
print(validate_mobile(mobile))
```

OUTPUT:

```
Enter Email Address: rohith@gmail.com
Enter Password: Roo@!@123
Enter Mobile Number: 1234567890
Valid Email
Strong Password
Invalid Mobile Number
```

2.

```
# DFA Simulator

# Input DFA details
states = input("Enter states (space separated): ").split()
alphabet = input("Enter input alphabet (space separated): ").split()

print("\nEnter Transition Table:")
transitions = {}

for state in states:
    transitions[state] = {}
    for symbol in alphabet:
        next_state = input(f" $\delta$ ({state}, {symbol}) = ")
        transitions[state][symbol] = next_state

initial_state = input("\nEnter Initial State: ")
final_states = input("Enter Final State(s) (space separated): ").split()

# Number of input strings
n = int(input("\nEnter number of input strings: "))

for i in range(n):
    string = input(f"\nEnter String {i+1}: ")

    current_state = initial_state
    path = [current_state]
    valid = True

    for ch in string:
        if ch not in alphabet:
            valid = False
            break
        current_state = transitions[current_state][ch]
        path.append(current_state)

    print("Transition Path:")
    print(" → ".join(path))

    if valid and current_state in final_states:
        print("Accepted")
    else:
        print("Rejected")
```

OUTPUT:

Enter states (space separated): q0 q1 q2

Enter input alphabet (space separated): a b

Enter Transition Table:

$\delta(q_0, a) = q_1$

$\delta(q_0, b) = q_0$

$\delta(q_1, a) = q_1$

$\delta(q_1, b) = q_2$

$\delta(q_2, a) = q_1$

$\delta(q_2, b) = q_0$

Enter Initial State: q0

Enter Final State(s) (space separated): q2

Enter number of input strings: 2

Enter String 1: abaab

Transition Path:

$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_1 \rightarrow q_1 \rightarrow q_2$

Accepted

Enter String 2: abb

Transition Path:

$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_0$

Rejected

3.

```
import re

# ----- Input Text -----
print("Enter the text (Type END on a new line to finish):")

lines = []
while True:
    line = input()
    if line.upper() == "END":
        break
    lines.append(line)

text = "\n".join(lines)

# ----- Menu -----
while True:
    print("\n----- Pattern Matching Menu -----")
    print("1. Search Date")
    print("2. Search Phone Number")
    print("3. Search Hashtag")
    print("4. Search Mention")
    print("5. Search Prefix")
    print("6. Search Suffix")
    print("7. Search Word")
    print("8. Exit")

    try:
        choice = int(input("Enter your choice: "))
    except ValueError:
        print("Please enter a number between 1 and 8.")
        continue

    if choice == 1:
        # Date Search (DD/MM/YYYY)
```

```
    dates = re.findall(r'\b\d{2}/\d{2}/\d{4}\b', text)

    if dates:
        print("Dates Found:", dates)
    else:
        print("No Date Found")

elif choice == 2:
    # Phone Number Search
    phones = re.findall(r'\b[6-9]\d{9}\b', text)

    if phones:
        print("Phone Numbers Found:", phones)
    else:
        print("No Phone Number Found")

elif choice == 3:
    # Hashtag Search
    hashtags = re.findall(r'#\w+', text)

    if hashtags:
        print("Hashtags Found:", hashtags)
    else:
        print("No Hashtag Found")

elif choice == 4:
    # Mention Search
    mentions = re.findall(r'@\w+', text)

    if mentions:
        print("Mentions Found:", mentions)
    else:
        print("No Mention Found")

elif choice == 5:
    # Prefix Search
    prefix = input("Enter Prefix: ")

    words = re.findall(r'\b' + re.escape(prefix) + r'\w*\b', text, re.IGNORECASE)

    if words:
        print("Matching Words:", words)
    else:
        print("No Matching Words Found")

elif choice == 6:
    # Suffix Search
    suffix = input("Enter Suffix: ")

    words = re.findall(r'\b\w*' + re.escape(suffix) + r'\b', text, re.IGNORECASE)

    if words:
```

```

        print("Matching Words:", words)
    else:
        print("No Matching Words Found")

elif choice == 7:
    # Word Search
    word = input("Enter Word: ")

    words = re.findall(r'\b' + re.escape(word) + r'\b', text, re.IGNORECASE)

    if words:
        print("Word Found:", words)
    else:
        print("Word Not Found")

elif choice == 8:
    print("Exiting Program...")
    break

else:
    print("Invalid Choice! Please enter a number between 1 and 8.")

```

OUTPUT:

Enter the text (Type END on a new line to finish):

Meeting on 12/09/2026

Call 9876543210

#NLP

@OpenAI

natural language processing

END

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: 1

Dates Found: ['12/09/2026']

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: 2

Phone Numbers Found: ['9876543210']

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: 3

Hashtags Found: ['#NLP']

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: 4

Mentions Found: ['@OpenAI']

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: 5

Enter Prefix: unhappy

No Matching Words Found

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: meeting

Please enter a number between 1 and 8.

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention

5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: 6

Enter Suffix: meeting

Matching Words: ['Meeting']

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: 7

Enter Word: Language

Word Found: ['language']

----- Pattern Matching Menu -----

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

Enter your choice: 8

Exiting Program...